

TinkerHub Tech Team - Project for Hardware

SENSOR MONITORING SYSTEM WITH LCD DISPLAY

1. Sensor Monitoring System with LCD Display

A sensor monitoring system is an electronic setup designed to continuously track environmental parameters such as temperature, light, humidity, or motion using sensors, and respond when certain conditions are met. In this project, the system uses a TMP36 temperature sensor and an LDR (light-dependent resistor) to measure temperature and light levels. The values are displayed on a 16×2 LCD screen in real time, allowing users to easily observe environmental changes. A threshold mechanism is implemented so that when the temperature exceeds a set value or the light level drops below a certain limit, an LED indicator is activated as an alert. This kind of system is widely applicable in smart homes, greenhouses, weather stations, and industrial monitoring, where real-time awareness and automatic responses to changing conditions are essential.

By simulating in Tinkercad, students and beginners can experiment with wiring, code, and logic without the risk of damaging real components, making it an excellent platform for learning how monitoring systems work in real-world applications.

2. Components Required

- **Arduino Uno R3** : This device has an ATmega328 Microcontroller, 6 analog inputs and 13 digital inputs/outputs.
- **TMP36 / Temperature Sensor** : This sensor measures Temperature. It contains semiconductor material that changes electrical properties consistently with changing temperatures. The center pin's signal changes as the temperature changes, and the microcontroller's program can convert this signal from 0 to 1023 to a temperature at °C or °F.
- **Photoresistor** : Variable Resistor that detects light. When light energy contracts the sensor, material absorbs some of the light energy and converts it to electrical energy, becoming less resistive.
- **LCD Display 16 x 2** : A Liquid Crystal Display (LCD) capable of displaying two lines of text, with 16 characters each.
- **Potentiometer** : A Variable Resistor controlled by a knob. It works by changing the position of a movable contact on a resistive material in relation to a stationary contact, resulting in a changing resistance as the contact is moved. **Here, we need 10 kΩ Potentiometer**
- **Resistor** : Limits the flow of Electric Current through the Circuit.
 - 10 Ω Resistor for LDR (Photoresistor) Voltage Divider

- 220 Ω Resistor(s) for LED LCD Backlight
- **LED** : An LED is a semiconductor material that emit light when current flows through them in the correct direction.
- **Breadboard**: For prototyping circuits.
- **Jumper Wires**

3. Circuit Design

- **Power Ground Connections** : Arduino Uno 5V pin supplies power to the TMP36 sensor, LDR, LCD module, and potentiometer. Arduino GND pins are connected to all components that require ground, including TMP36, LDR resistor, LCD, potentiometer, and LED cathodes. All GND points are common, meaning they are internally connected inside the Arduino board. You can safely use either GND pin or extend GND through a breadboard rail.
- **TMP36 / Temperature Sensor** : The TMP36 has three pins.
 - Left pin (VCC) $\rightarrow$ +5V
 - Middle pin (VOUT) $\rightarrow$ Arduino A0 (analog input)
 - Right pin (GND) $\rightarrow$ GND

The sensor outputs a voltage proportional to the temperature. At 25°C, it gives 0.75V, increasing by 10mV per °C.

- **LDR (Light Sensor)** : LDR is connected as a voltage divider with a fixed 10k resistor.
 - One leg of LDR $\rightarrow$ +5V
 - Other leg of LDR $\rightarrow$ Arduino A1 and $\rightarrow$ one side of a 10 k Ω resistor
 - Other side of 10 k Ω resistor $\rightarrow$ GND

As light increases, LDR resistance decreases, changing the voltage at A1. Arduino reads this value (0–1023).

- **LED Indicator** : LED is used as an alert when thresholds are crossed.
 - Anode (long leg) $\rightarrow$ Arduino Pin 8 through a 220 Ω resistor
 - Cathode (short leg) $\rightarrow$ GND
 - Other side of 10 k Ω resistor $\rightarrow$ GND

The resistor limits current, protecting the LED from burning out.

- **LCD 16 $\times$ 2 Display** :
 - Potentiometer Wiring for Contrast

LCD Pin	Name	Connection
1	VSS	GND
2	VD	+5V
3	VO	Wiper of Potentiometer
4	RS	PIN 7
5	RW	GND
6	E	PIN 6
7	D0	-
8	D1	-
9	D2	-
10	D3	-
11	D4	PIN 5
12	D5	PIN 4
13	D6	PIN 3
14	D7	PIN 2
15	LED (Anode)	+5V via 220 Resistor
16	LED (Cathode)	GND

Table 1: Connections of LCD Display

- * One outer pin $\rightarrow$ 5V
- * Other outer pin $\rightarrow$ GND
- * Middle (wiper) $\rightarrow$ VO (LCD pin 3)
- Backlight Wiring
 - * Pin 15 (Anode) $\rightarrow$ 5V via 220 resistor
 - * Pin 16 (Cathode) $\rightarrow$ GND

4. Design Summary

- The TMP36 measures temperature and sends analog voltage to A0.
- The LDR circuit provides a light-dependent voltage to A1.
- The Arduino reads these values, displays them on the LCD, and turns the LED on when *temperature* $> 30^{\circ}\text{C}$ or *light* $< \text{threshold}$.
- LCD runs in 4-bit mode to save Arduino pins, controlled by pins 7–2.
- Potentiometer allows adjusting display contrast.
- Always use resistors for LED and backlight to limit current.
- Adjust the potentiometer until text is visible on the LCD screen.
- In Tinkercad, you can use sliders to change light intensity on the LDR and temperature on the TMP36 to test the thresholds.

5. Working

Outline the proposed stages of your project:

- **TMP36 Sensor** : Outputs a voltage proportional to temperature. Arduino converts this to °C.
- **LDR Sensor** : Forms a voltage divider with a 10k resistor; Arduino reads the brightness level.
- **LCD Display** : Shows live temperature and light readings.
- **LED Alert** : Turns ON if
 1. *Temperature* > 30 °C (or)
 2. *Lightlevel* < 500.

6. Simulation Testing

- Run the simulation in Tinkercad or upload to a real Arduino Uno.
- Open the Serial Monitor to see live readings. LCD will also display readings.
- The LED should turn ON when conditions are exceeded.

7. Arduino Code

```
#include <LiquidCrystal.h>

// LCD pins: RS=7, EN=6, D4=5, D5=4, D6=3, D7=2
LiquidCrystal lcd(7, 6, 5, 4, 3, 2);

// Sensors and LED
#define LED_PIN 8
#define TMP_PIN A0
#define LDR_PIN A1

// Thresholds
float tempThreshold = 30.0; // Temperature in °C
int lightThreshold = 500;   // Light level (0-1023)

void setup() {
  Serial.begin(9600);
  pinMode(LED_PIN, OUTPUT);

  // Initialize LCD
  lcd.begin(16, 2);
```

```
    lcd.setCursor(0, 0);
    lcd.print("System Ready...");
    delay(1000);
    lcd.clear();
}

void loop() {
    // --- TMP36 Temperature ---
    int tempReading = analogRead(TMP_PIN);
    float voltage = tempReading * (5.0 / 1023.0);
    float temperatureC = (voltage - 0.5) * 100.0;

    // --- LDR Light ---
    int lightValue = analogRead(LDR_PIN);

    // --- LED Logic ---
    if (temperatureC > tempThreshold || lightValue < lightThreshold) {
        digitalWrite(LED_PIN, HIGH); // ON
    } else {
        digitalWrite(LED_PIN, LOW); // OFF
    }

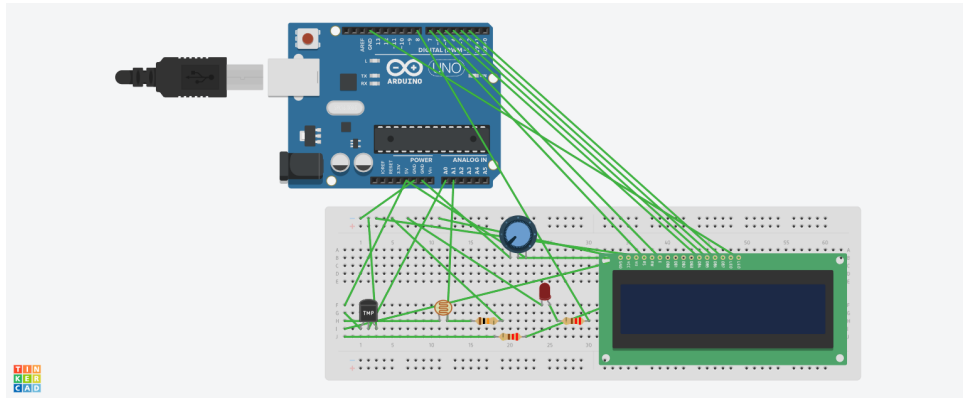
    // --- Serial Monitor Output ---
    Serial.print("Temperature: ");
    Serial.print(temperatureC);
    Serial.print(" °C, Light: ");
    Serial.println(lightValue);

    // --- LCD Display ---
    lcd.setCursor(0, 0);
    lcd.print("Temp: ");
    lcd.print(temperatureC, 1);
    lcd.print((char)223); // Degree symbol
    lcd.print("C  "); // Spaces to clear old digits

    lcd.setCursor(0, 1);
    lcd.print("Light: ");
    lcd.print(lightValue);
    lcd.print("  "); // Spaces to clear old digits

    delay(1000); // Update every second
}
```

8. Schematic Diagram



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